**Nptel Online Certification Course**
**Indian Institute of Technology Kharagpur**
**Computer Vision**
**Assignment - Week 4**

**Number of questions: 10** **Total marks: 10x2=20**

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**QUESTION 1:** **Type: MCQ**

Given the right epipole $e'(1, 4)$ and a point $x'(3, 2)$ in right image plane of a stereo imaging set up, compute the epipolar line $l'$ containing the point $x'$, in right image plane in homogeneous coordinate system.

a) $(0.2, 0.2, 1)$

b) $(-0.2, 0.2, 1)$

c) $(-0.2, -0.2, 1)$

d) $(0.2, -0.2, 1)$

**Correct Answer:** : c)
**Detailed Solution:** Epipolar line $l' = e' \times x'$ in homogeneous coordinate system.

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**QUESTION 2:** **Type: MCQ**

Assume that a stereo imaging setup has two image planes, left and right. Which of the following could be a possible right null vector (or left epipole for the corresponding stereo imaging setup) of the given fundamental matrix $F = \begin{bmatrix} 20 & 12 & 0 \\ 8 & -7 & 59 \\ -4 & 33 & -177 \end{bmatrix}$.

a) $(3, 5, 1)$

b) $(-6, 10, 2)$

c) $(6, -10, 1)$

d) $(3, -5, 1)$

**Correct Answer:** : b)

**Detailed Solution:** Check for $Fe = 0$, where $e$ is the left epipole or right null vector of $F$.

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**FOR QUESTIONS 3, 4 AND 5:**

Consider a stereo imaging set up camera with left calibration matrix $K = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and a right calibration matrix $K' = \begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$. The viewing direction of right camera is $R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

Based on the given parameters, solve the following question 3, 4 and 5:

**QUESTION 3:** **Type: Comprehensive**

Find the homography matrix $H$.

a) $\begin{bmatrix} 1 & 1 & 3 \\ 3 & 1 & 1 \\ 2 & 0 & 0 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 5 & 0 & 3 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 5 & 3 \\ 2 & 1 & 2 \\ 3 & 3 & 2 \end{bmatrix}$

d) $\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

**Correct Answer:**d)

**Detailed Solution:**

Homography is given by $H = K'RK^{-1}$.

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**QUESTION 4:** **Type: Comprehensive**

Given translation vector $t$ as $(2, 1, 0)$ of the center of the right camera with reference to that of the left camera, compute the right epipole.

a) $(4, 2, 1)$

b) $(7, 1, 0)$

c) $(7, 1, 2)$

d) $(2, 1, 0)$

**Correct Answer:** b)

**Detailed Solution:**

Right epipole is given by $e' = K't$.

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**QUESTION 5:** **Type: Comprehensive**

Compute the fundamental matrix $F$ assuming the translation vector of the center of the right camera with reference to the left camera is $(2, 1, 0)$.

a) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & -7 & 0 \\ -1 & 0 & 4 \end{bmatrix}$

b) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & -7 & 0 \\ -1 & 6 & 4 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ -7 & -6 & -20 \end{bmatrix}$

d) $\begin{bmatrix} 0 & -1 & 0 \\ 0 & 7 & 0 \\ 1 & -6 & -4 \end{bmatrix}$

**Correct Answer:** b)

**Detailed Solution:**

Fundamental matrix is given by $F = [e']_X H$.

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**FOR QUESTIONS 6 AND 7:**

Given fundamental matrix $F = \begin{bmatrix} -1 & 3 & 2 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

Based on the given data solve the following questions 5 and 6:

**QUESTION 6:** **Type: Comprehensive**

Given two points $(2, 3)$ and $(4, -2)$ in the left image. Compute the epipolar lines corresponding to them in the right image plane.

a) $(-9, 5, 5), (-8, 2, 2)$

b) $(9, 5, -5), (-8, 2, 2)$

c) $(9, 5, 5), (-8, 2, 2)$

d) $(-9, 5, 5), (8, 2, 2)$

**Correct Answer:** : c)

**Detailed Solution:** Epipolar line is given by $l' = Fx$

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**QUESTION 7:** **Type: Comprehensive**

Find the right epipole.

a) $(0, 58, 58)$

b) $(0, -58, 58)$

c) $(0, 58, -58)$

d) $(0, -58, -58)$

**Correct Answer:** : b)
**Detailed Solution:** Epipolar line is given by $e' = l'_1 \times l'_2$

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**QUESTION 8:** **Type: Numeric**

Consider a stereo imaging set up with two cameras $P = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ (left camera) and $P' = \begin{bmatrix} 2 & 0 & 0 & 5 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ (right camera). If the image coordinates of a 3-D point are $(10, 3)$ and $(15, 3)$ in left and right cameras, compute its depth (z-coordinate) in the 3D. (Answer only the magnitude of the depth)

**Correct Answer:** 1

**Detailed Solution:**

The depth is given by $Z = \frac{Kt}{x'-x} = \frac{2\times\frac{5}{2}}{15-10}$.

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**QUESTION 9:** **Type: MCQ**

Given the rotational (R) and translation (t) transformations of the viewing direction and camera center of right camera $R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ and $t$ as $(1, 1, 1)$, compute the essential matrix $E$ of a calibrated stereo set-up.

a) $\begin{bmatrix} 0 & 2 & 1 \\ -1 & 1 & 0 \\ -2 & 0 & -1 \end{bmatrix}$

b) $\begin{bmatrix} 0 & -2 & -1 \\ 1 & -1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$

c) $\begin{bmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

d) $\begin{bmatrix} -2 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

**Correct Answer:** c)

**Detailed Solution:**

Essential matrix is given by $E = [t']_X R$.

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**QUESTION 10:** **Type: MSQ**

Consider a stereo imaging set up with two cameras $P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ (left camera) and $P' = \begin{bmatrix} 1 & 2 & 5 & 1 \\ 2 & 0 & 0 & 2 \\ 3 & 0 & 2 & 1 \end{bmatrix}$ (right camera), which of the following options are correct?

a) $F = \begin{bmatrix} 4 & 0 & 0 \\ 2 & 2 & -3 \\ 0 & -4 & -10 \end{bmatrix}$

b) The right epipole is $(1, 2, 1)$.

c) $F = \begin{bmatrix} 4 & 0 & 4 \\ -2 & 2 & 3 \\ 0 & -4 & -10 \end{bmatrix}$

d) The right epipole is $(0, 0, 1)$.

**Correct Answer:** : b), c)
**Detailed Solution:** Right epipole will be $e' = P'C$. Fundamental matrix $F = [m]_X M$, where $m$ is the last column of $P'$ and $M$ is the left 3x3 matrix of $P'$.

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